Kanyisola Aminat Fagbayi

Data Scientist | AI & Machine Learning Engineer | Public Sector Digital Transformation & MEL

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PROFILE

Data scientist and AI engineer with seven years inside the Ogun State Government, where I design monitoring for social programmes and turn programme data into decisions for senior officials. I build machine learning models, AI applications and data platforms, from a maternal health risk model to retrieval-based assistants, fraud and complaint routing systems, and payment infrastructure. My work sits where data, AI and public service meet: using evidence and technology to help programmes reach the people they are meant for. MSc Data Science (University of East London) and alumna of the Bosun Tijani Foundation AI Engineering Fellowship with the Nigerian Communications Commission.

CORE SKILLS

- **Data science and machine learning:** Python (pandas, NumPy, scikit-learn), R, SQL; classification and regression, Random Forest, XGBoost, logistic regression; feature engineering, cross-validation, class imbalance (SMOTE), calibration, SHAP explainability
- **AI engineering:** retrieval-augmented generation (RAG), LangGraph multi-agent systems, OpenAI and Gemini APIs, prompt engineering, LLM evaluation; computer vision (ResNet50) and OCR (PaddleOCR)
- **Engineering and deployment:** FastAPI, REST APIs and webhooks, PostgreSQL, Docker, Google Cloud, Git/GitHub, automated testing, Next.js (working knowledge)
- **Monitoring, evaluation and learning:** indicator frameworks, monitoring plans, beneficiary tracking, disaggregated reporting, data quality checks, dashboards (Power BI), donor and partner reporting
- **Public sector delivery:** evidence packs and executive briefings, multi-agency coordination, training field teams, programme oversight

EXPERIENCE

Special Assistant to the Governor, Office of the Deputy Governor, Ogun State Government, Abeokuta — Oct 2019 to present

- Designed and run monitoring and evaluation for the Oko'wo Dapo interest-free loan scheme for women: indicators for eligibility, disbursement, follow-up and outcomes, beneficiary records, and disaggregated reports across local government areas.
- Supported the Deputy Governor as Chair of the Nigeria for Women Project State Steering Committee (World Bank-assisted): prepared meeting briefs, tracked action points and reviewed progress reports from the implementation unit.
- Led production and distribution for a five-month COVID-19 food relief programme (2020) and relief during the fuel subsidy removal, managing logistics and relief budgets, with impact monitoring and beneficiary feedback reported to government and NGO partners; coordinated information flow for the State COVID-19 Task Force.
- Produce dashboards, evidence packs and briefings that turn programme data into decisions for senior officials, including briefing notes for national meetings such as the National Economic Council.
- Supported a state-wide review of the roles of government agencies, compiling submissions and identifying overlapping mandates.
- Introduced data and digital tools in the office to strengthen programme oversight and reporting; train and coordinate field teams for data collection.

Co-founder & Lead Engineer, Kollectors *(levy and dues collection platform, in pre-launch testing)* — Jul 2026 to present

- Co-founded and lead engineering of a payments platform for residential estates and cooperative societies, built with FastAPI, Next.js and PostgreSQL (Supabase) on Nomba payment APIs.

- Designed automatic reconciliation: each household gets a dedicated virtual account, and payments are confirmed only after a signed webhook and server-side verification.
- Ran end-to-end testing and fixed data-integrity defects so removed members drop out of billing, reminders and reports; built a treasurer dashboard, transparency reports and WhatsApp payment reminders.

Technical Assistant to the MD/CEO, Glenwood Property Development Co. Ltd., Lagos 2016 to 2019

- Ran feasibility studies, competitor analysis and financial assessments for real estate investments; validated payment requests against Bills of Quantities; liaised with government officials and regulators.

Polio Vaccination Monitor, Hanwa, Zaria, Kaduna State *(with the Kaduna State Government, Gates Foundation and UNICEF)* 2015 to 2016

- Monitored house-to-house oral polio vaccination rounds, collected field data, verified coverage and reported programme reach.

SELECTED PROJECTS

MamaDey AI: early maternal risk prediction *(MSc dissertation, completed)* github.com/Cunyie08/mamadey_ai

- Engaged with the Ogun State Ministry of Health on why maternal deaths remain high despite free antenatal care, BHCPF and MAMII: interventions are largely facility-based, many women leave care early, and paper records leave home births out of the data.
- Built and compared Logistic Regression, Random Forest and XGBoost on Nigeria DHS 2018 data for 2,563 women (AUC-ROC 0.81 to 0.86); recommended the model that identified the most high-risk women and used SHAP to explain the main risk drivers.
- Designed a proof-of-concept framework linking community health worker data to alerts and to facility and ministry dashboards connected to DHIS2.

TaxPyre Nigeria: plain-language tax law assistant *(team project)*

- Agentic RAG system over 500+ pages of Nigerian tax reform legislation, with conditional retrieval, multi-turn memory and mandatory citations to document, section and page.

SENTINNEL Banking: multi-agent complaint and fraud intelligence *(capstone, nine-person team; architect and orchestrator)*

- LangGraph system that routes customer complaints, flags potential fraud with a Random Forest classifier trained on synthetic Nigerian banking data, grounds decisions in bank policy through retrieval, and keeps a PostgreSQL audit trail; evaluation harness with an LLM judge.

SabiLens: counterfeit product detection for regulators *(team capstone, deployed on Google Cloud)*

- Decision engine combining ResNet50 image analysis, PaddleOCR label reading and regulatory database checks; FastAPI backend with 94 automated tests and tamper-evident evidence packages for regulators such as NAFDAC.

AgroUSSD: connecting farmers directly to buyers *(prototype, AI Engineering Fellowship)*

- Python prototype simulating a USSD service for basic phones: farmers register, record harvests, compare market prices and connect directly with buyers; designed for local-language support and a live short code.

EDUCATION

MSc Data Science, University of East London *(dissertation completed; award pending conferment)* 2024 to 2026

AI Engineering Fellowship, Bosun Tijani Foundation and Nigerian Communications Commission (first cohort) 2025 to 2026

BSc Mechanical Engineering, University of Lagos 2008 to 2014

ADDITIONAL

Nigerian national. Languages: English (fluent), Yoruba (fluent). References available on request.